TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

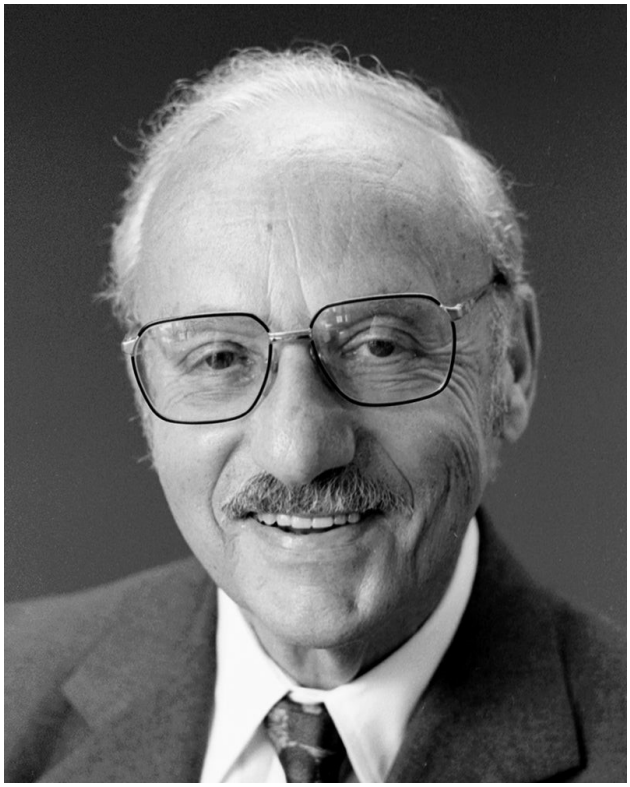


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

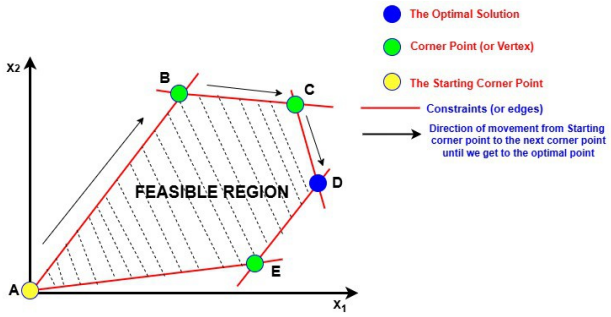


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Problema 4
- Amount of decision variables: 2
- Amount of restrictions: 3
- Total Amount of variables (decision + slack): 5
- Type of problem: Maximization

Equations

$$
\begin{aligned}
&\text{Max } Z = 5\,x_1 + 3\,x_2 \\
&\text{subject to:} \\
&\qquad 4\,x_1 + 2\,x_2 \leq 12 \\
&\qquad 4\,x_1 + x_2 \leq 10 \\
&\qquad x_1 + x_2 \leq 4 \\
&\qquad x_1, x_2 \geq 0
\end{aligned}
$$

Initial Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	-5.0000	-3.0000	0.0000	0.0000	0.0000	0.0000
0.0000	4.0000	2.0000	1.0000	0.0000	0.0000	12.0000
0.0000	4.0000	1.0000	0.0000	1.0000	0.0000	10.0000
0.0000	1.0000	1.0000	0.0000	0.0000	1.0000	4.0000

Intermediate Tables

Iteration 1

Entering variable: x_1 (most negative coefficient in the Z row: -5.0000). Leaving variable: s_2 (row 2, minimum ratio 2.5000).

Finding the Pivot

Z	x_1	x_2	s_1	s_2	s_3	b	P
1.0000	-5.0000	-3.0000	0.0000	0.0000	0.0000	0.0000	-
0.0000	4.0000	2.0000	1.0000	0.0000	0.0000	12.0000	3.0000
0.0000	**4.0000**	1.0000	0.0000	1.0000	0.0000	10.0000	2.5000
0.0000	1.0000	1.0000	0.0000	0.0000	1.0000	4.0000	4.0000

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	-1.7500	0.0000	1.2500	0.0000	12.5000
0.0000	0.0000	1.0000	1.0000	-1.0000	0.0000	2.0000
0.0000	1.0000	0.2500	0.0000	0.2500	0.0000	2.5000
0.0000	0.0000	0.7500	0.0000	-0.2500	1.0000	1.5000

Iteration 2

Entering variable: x_2 (most negative coefficient in the Z row: -1.7500). Leaving variable: s_1 (row 1, minimum ratio 2.0000).

Tie: 2 rows share the minimum ratio 2.0000. The tie was resolved by choosing the topmost row. The next table will have a basic variable equal to zero (degenerate solution).

Finding the Pivot

Z	x_1	x_2	s_1	s_2	s_3	b	P
1.0000	0.0000	-1.7500	0.0000	1.2500	0.0000	12.5000	-
0.0000	0.0000	**1.0000**	1.0000	-1.0000	0.0000	2.0000	2.0000
0.0000	1.0000	0.2500	0.0000	0.2500	0.0000	2.5000	10.0000
0.0000	0.0000	0.7500	0.0000	-0.2500	1.0000	1.5000	2.0000

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	1.7500	-0.5000	0.0000	16.0000
0.0000	0.0000	1.0000	1.0000	-1.0000	0.0000	2.0000
0.0000	1.0000	0.0000	-0.2500	0.5000	0.0000	2.0000
0.0000	0.0000	0.0000	-0.7500	0.5000	1.0000	0.0000

Iteration 3

Entering variable: s_2 (most negative coefficient in the Z row: -0.5000). Leaving variable: s_3 (row 3, minimum ratio 0.0000).

Degenerate pivot: the minimum ratio is zero, so this iteration changes the basis but not the value of Z.

Finding the Pivot

Z	x_1	x_2	s_1	s_2	s_3	b	P
1.0000	0.0000	0.0000	1.7500	-0.5000	0.0000	16.0000	-
0.0000	0.0000	1.0000	1.0000	-1.0000	0.0000	2.0000	-
0.0000	1.0000	0.0000	-0.2500	0.5000	0.0000	2.0000	4.0000
0.0000	0.0000	0.0000	-0.7500	**0.5000**	1.0000	0.0000	0.0000

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	1.0000	0.0000	1.0000	16.0000
0.0000	0.0000	1.0000	-0.5000	0.0000	2.0000	2.0000
0.0000	1.0000	0.0000	0.5000	0.0000	-1.0000	2.0000
0.0000	0.0000	0.0000	-1.5000	1.0000	2.0000	0.0000

Final Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	0.0000	0.0000	1.0000	0.0000	1.0000	16.0000
0.0000	0.0000	1.0000	-0.5000	0.0000	2.0000	2.0000
0.0000	1.0000	0.0000	0.5000	0.0000	-1.0000	2.0000
0.0000	0.0000	0.0000	-1.5000	1.0000	2.0000	0.0000

Solution

Solution

The optimal solution is:

- $x_1 = 2.0000$
- $x_2 = 2.0000$
- $s_1 = 0.0000$ (non-basic)
- $s_2 = 0.0000$
- $s_3 = 0.0000$ (non-basic)

The optimal value of the objective function is: $Z = 16.0000$

Degenerate problem

During the execution (or in the final table) at least one basic variable took the value zero, so the problem is degenerate: more constraints than necessary meet at the vertex. Ties in the minimum ratio test were resolved by choosing the topmost row (smallest row index); ties for the entering column were resolved by choosing the leftmost column. The algorithm did not cycle.